Math 133: Quiz 4

Friday, October 9, 2026 25 minutes 10 points

Name: _____

Show your work. Give exact answers. Justify direct substitution briefly and show any algebraic simplification. Use $\pm\infty$ for unbounded behavior and DNE otherwise when no limit exists. A simple scientific calculator is allowed but not needed.

1. (4 points) Evaluate the limits.

(a) $\lim_{x\to 0} \dfrac{1+3^x}{x^2+2}$.

(b) $\lim_{x\to 4} \dfrac{x^2-x-12}{x-4}$.

(c) $\lim_{x\to+\infty} \left(\sqrt{x^2+6x}-x\right)$.

(d) Find the left-hand, right-hand, and two-sided limits at $x=-1$ of $\dfrac{1}{x+1}$.

2. (2 points) Find k so that H is continuous at $x=2$.

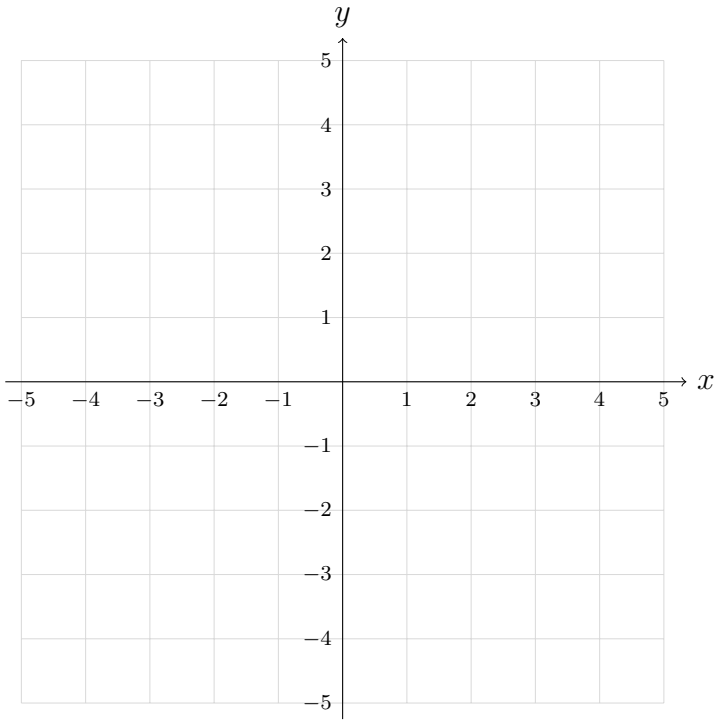
$$H(x)=\begin{cases} kx-1, & x\le 2,\\ x+5, & x>2. \end{cases}$$

$k =$ _____

3. (4 points) Let

$$g(x) = \begin{cases} x+1, & x < 1, \\ 0, & x = 1, \\ 3-x, & 1 < x < 3, \\ x-2, & x \geq 3. \end{cases}$$

(a) Graph g. Clearly mark open and filled endpoints.



(b) Complete the table.

a	$g(a)$	$\lim_{x \to a^-} g(x)$	$\lim_{x \to a^+} g(x)$	$\lim_{x \to a} g(x)$
1				
3				

(c) At each of $x = 1$ and $x = 3$, decide whether g is continuous. If not, classify the discontinuity as removable or a jump.

At $x = 1$:	At $x = 3$: